Databases: Reading Data with SELECT

Ask a database a question and get exactly the rows you want.

Databases | Beginner to Intermediate | Ages 12–18

InfiBee Coders

The Big Idea

- A database stores information in tables, like a spreadsheet with rows and columns.
- SQL is the language you use to talk to a database.
- `SELECT` reads data. It never changes anything, so it is safe to practise.
- Every query follows the order: `SELECT` columns, `FROM` a table, then optional extras.
- These examples use SQLite, and the same ideas work in MySQL and other databases.

The Students Table

- Our sample table is called `students`.
- Columns: `id`, `name`, `age`, `class`, `score`.
- Each row is one student.
- Text is written in single quotes, like `'6A'`. Numbers have no quotes.

Useful Clauses

- `WHERE` keeps only rows that match a condition.
- `AND`, `OR` and `NOT` combine conditions.
- `LIKE 'A%'` finds text that starts with A. `%` means any characters.
- `ORDER BY score DESC` sorts from high to low. `ASC` is low to high.
- `LIMIT 3` returns only the first 3 rows, and `DISTINCT` removes duplicates.
- `AS` gives a column a nicer name.

💻 Code Examples

See everything SQL

```sql
SELECT * FROM students;
```

▶ **Output:**

```
id  name   age  class  score
--  -----  ---  -----  -----
1   Aarav  12   6A     88
2   Mia    13   7B     95
3   Zoya   12   6A     72
4   Kabir  14   8C     64
5   Anaya  13   7B     81
6   Dev    12   6B     90
```

The star means all columns.

Pick columns SQL

```sql
SELECT name, score FROM students;
```

▶ **Output:**

```
name   score
-----  -----
Aarav  88
Mia    95
Zoya   72
Kabir  64
Anaya  81
Dev    90
```

Filter with WHERE SQL

```sql
SELECT name, score FROM students WHERE score > 85;
```

▶ **Output:**

```
name   score
-----  -----
Aarav  88
Mia    95
Dev    90
```

Combine conditions SQL

```sql
SELECT name, age FROM students WHERE age = 12 AND score >= 80;
```

▶ **Output:**

```
name   age
-----  ---
Aarav  12
Dev    12
```

Search with LIKE SQL

```sql
SELECT name FROM students WHERE name LIKE 'A%';
```

▶ **Output:**

```
name
-----
Aarav
Anaya
```

Matches names that start with A.

Sort and limit SQL

```sql
SELECT name, score FROM students ORDER BY score DESC LIMIT 3;
```

▶ **Output:**

```
name   score
-----  -----
Mia    95
Dev    90
Aarav  88
```

The top 3 scores.

Unique values and nicer names SQL

```sql
SELECT DISTINCT class AS classroom FROM students;
```

▶ **Output:**

```
classroom
---------
6A
7B
8C
6B
```

📚 Key Terms

Database
An organised collection of data.

Table
Rows and columns of data.

Row
One record, like one student.

Column
One kind of data, like name.

SQL
The language for databases.

Query
A question you ask a database.

SELECT
Reads data.

WHERE
Filters rows.

ORDER BY
Sorts the results.

LIMIT
Restricts the number of rows.

🧮 Quick Reference

```
SELECT * FROM table;
SELECT col1, col2 FROM table;
... WHERE col > 10 AND col2 = 'x'
... WHERE name LIKE 'A%'
... ORDER BY col DESC
... LIMIT 5
SELECT DISTINCT col FROM table;
```

✏️ Worked Examples

Q: List the names of students with a score above 85.
A: SELECT name FROM students WHERE score > 85; gives Aarav, Mia and Dev.

Q: Find the highest scoring student.
A: SELECT name, score FROM students ORDER BY score DESC LIMIT 1; gives Mia with 95.

Q: Show only the age 13 students.
A: SELECT name FROM students WHERE age = 13; gives Mia and Anaya.

🚫 Common Mistakes

- Forgetting the semicolon at the end of a query.
- Using double quotes for text. Use single quotes: `'6A'`.
- Writing `WHERE` after `ORDER BY`. The order is `WHERE` first.

- Using `=` with `LIKE` patterns. Use `LIKE 'A%'`.
- Using `SELECT *` on huge tables when you only need a few columns.

👨‍👩‍👧 Parent Corner

- Compare a table to a class register: each row is a child, each column is a detail.
- Ask your child to write a question in English first, then turn it into SQL.
- Try a free tool like DB Browser for SQLite, or an online SQL playground, to practise.
- SELECT is safe. It only reads and cannot damage the data.
- Ask them what they would store in a database: pets, scores, books or games.

🎯 Quick Check

1. Which command reads data?
✅ SELECT.

2. What does SELECT * mean?
✅ Select all columns.

3. Which clause filters rows?
✅ WHERE.

4. How do you sort from highest to lowest?
✅ ORDER BY column DESC.

5. How do you return only 3 rows?
✅ LIMIT 3.